I- Relations & Functions

Investigation

In some countries it is customary at business meetings to shake hands with everybody in the meeting. If there are 2 people there is 1 handshake, if there are 3 people there are 3 handshakes and so on.

1. How many handshakes are there for 4 people ?
2. Copy and complete this table :

Number of people	Number of handshakes
2	
3	
4	
5	
6	
7	
8	
9	
10	

3. Plot the points on a Cartesian coordinate plane with the number of people on the x-axis and the number of handshakes on the y-axis.
4. Write a formula for the number of handshakes, H, in terms of the number of people, n.

$$H = \frac{n(n-1)}{2}$$

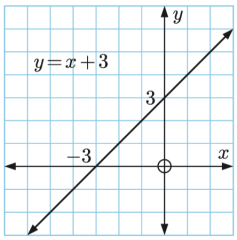
Definition

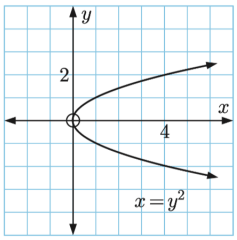
A relation is any set of points which connect two variables.

A relation is often expressed in the form of an equation connecting the variables x and y. In this case the relation is a set of points (x, y) in the Cartesian plane.

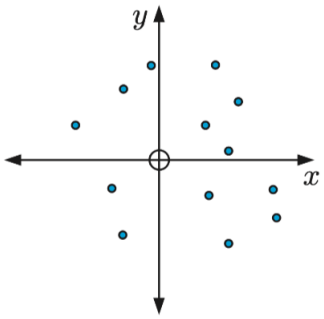
Example

For example, $y = x + 3$ and $x = y^2$ are the equations of two relations. Each equation generates a set of ordered pairs, which we can graph :





Remark. A relation may not be able to be defined by an equation.



These 13 points form a relation.

Definition

A function, sometimes called a mapping, is a relation in which no two different ordered pairs have the same x-coordinate or first component.

Remark. Every function is a relation, but not every relation is a function.

Property

Using Algebra :

If a relation is given as an equation, and the substitution of any value for x results in one and only one value of y, then the relation is a function.

Example

$y = 3x - 1$ is a function, since each x has only one y.

$x = y^2$ is not a function, since if $x = 4$, then $y = \pm 2$.

Property

Using Geometry :

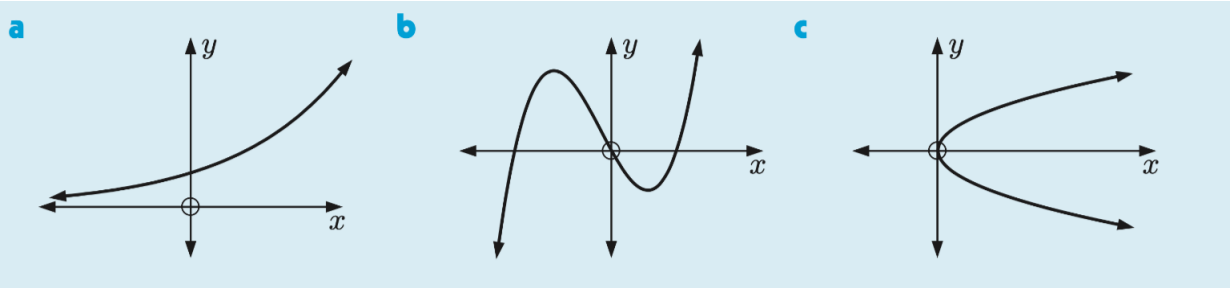
If we draw all possible vertical lines on the graph of a relation, the relation :

— is a function if each line cuts the graph no more than once ;

— is not a function if at least one line cuts the graph more than once.

Example

Which of the following relations are functions ?



Remark. Graphical note :

— If a graph contains a small open circle such as ⟜, this point is not included.

— If a graph contains a small filled-in circle such as ⟞, this point is included.

— If a graph contains an arrow head at an end such as ⟶, then the graph continues indefinitely in that general direction, or the shape may repeat as it has done previously.

II- Some vocabulary

a) Notation

Definition

$f(x)$ is the value of y for a given value of x, so $y = f(x)$.

Definition

A linear function is a function of the form

$$f(x) = ax + b$$

where a, b are real constants. The graph of a linear function is a straight line.

Example

If $f(x) = 5 - x - x^2$, find in simplest form :

1. $f(-x)$
2. $f(x+2)$

b) Domain and range

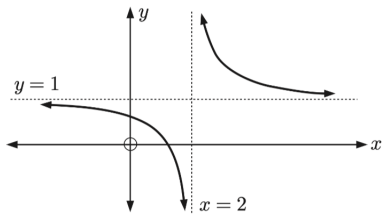
Definition

— The domain of a relation is the set of values of x in the relation.

— The range of a relation is the set of values of y in the relation.

The domain and range of a relation are often described using set notation.

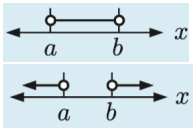
Example



x can take all values except 2, so the domain is $\{x \mid x \neq 2\}$.

y can take all values except 1, so the range is $\{y \mid y \neq 1\}$.

Definition



For numbers *between* a and b we write $a < x < b$.

For numbers *outside* a and b we write $x < a$ or $x > b$.

Remark. To fully describe a function, we need both a rule and a domain.

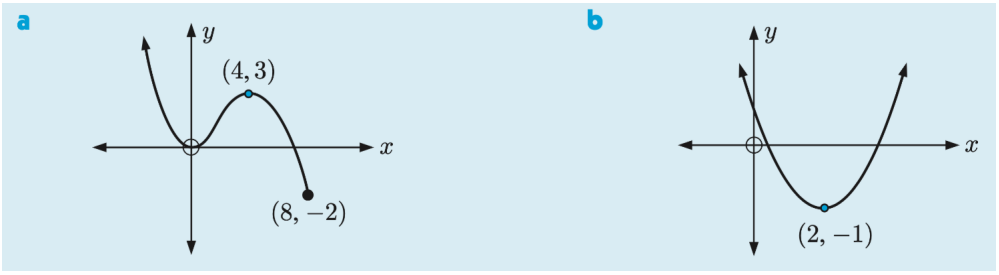
If a domain is not specified, we use the natural domain, which is the largest part of $\mathbb{R}$ for which $f(x)$ is defined.

Example

$f(x)$	**Natural domain**
x^2	$x \in \mathbb{R}$
$\sqrt{x}$	$x \geqslant 0$
$\frac{1}{x}$	$x \neq 0$
$\frac{1}{\sqrt{x}}$	$x > 0$

Example

For each of the following graphs, state the domain and range.



Example

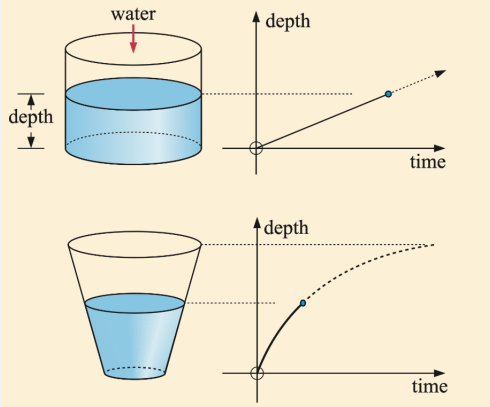
State the domain and range of each of the following functions :

1. $f(x) = \sqrt{x-5}$
2. $f(x) = \dfrac{1}{x-5}$
3. $f(x) = \dfrac{1}{\sqrt{x-5}}$

Investigation

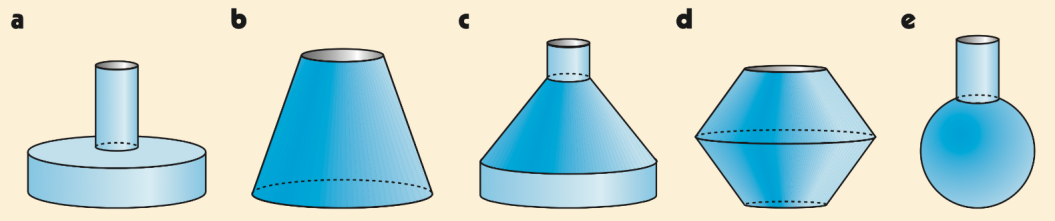
When water is added at a constant rate to a cylindrical container, the depth of water in the container is a linear function of time. This is because the volume of water added is directly proportional to the time taken to add it. If the water was not added at a constant rate, the depth of water over time would not be a linear function.

The linear depth–time graph for a cylindrical container is shown below.



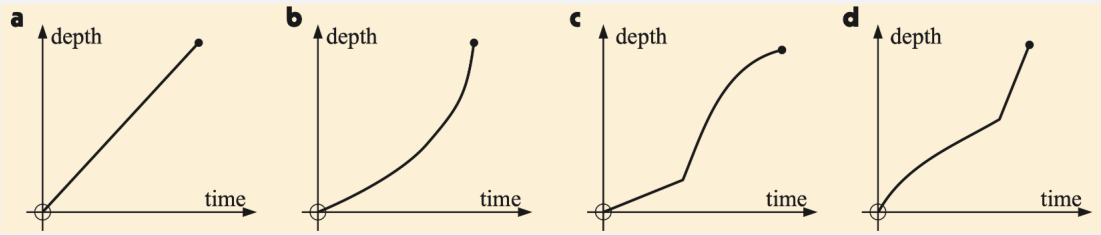
In this investigation, we explore the changes in the graph for different shaped containers such as the conical vase.

1. By examining the shape of each container, predict the depth–time graph when water is added at a constant rate.



2. Write a brief report on the connection between the shape of a vessel and the corresponding shape of its depth–time graph. First examine cylindrical containers, then conical, then other shapes.Gradients of curves must be included in your report.

3. Suggest containers which would have the following depth–time graphs :



III- Composite functions

Definition

Given $f : x \mapsto f(x)$ and $g : x \mapsto g(x)$, the composite function of f and g will convert x into $f(g(x))$.

$f \circ g$ is used to represent the composite function of f and g. It means "f following g".

$$(f \circ g)(x) = f(g(x)) \quad \text{or} \quad f \circ g : x \mapsto f(g(x)).$$

Remark. In general, $(f \circ g)(x) \neq (g \circ f)(x)$.

Example

Given $f : x \mapsto 2x + 1$ and $g : x \mapsto 3 - 4x$, find in simplest form :

1. $(f \circ g)(x)$
2. $(g \circ f)(x)$

Example

Given $f(x) = 6x - 5$ and $g(x) = x^2 + x$, determine :

1. $(g \circ f)(-1)$
2. $(f \circ f)(0)$

Remark. Consider $f(x) = x^2$ with domain $x \in \mathbb{R}$ and $g(x) = \sqrt{x}$ with domain $x \geqslant 0$.

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= (\sqrt{x})^2 \\ &= x\end{aligned}$$

The domain of $(f \circ g)(x)$ is $x \geqslant 0$, not $\mathbb{R}$, since $(f \circ g)(x)$ is defined using function $g(x)$.

IV- Sign diagram

For the function $f(x)$, the sign diagram consists of :

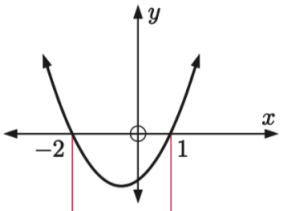
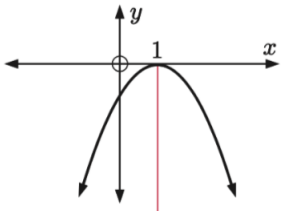
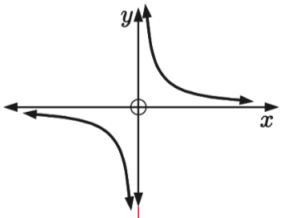
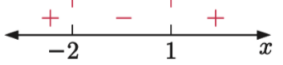
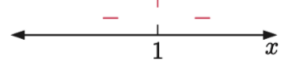
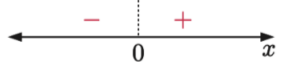
— a horizontal line which is really the x-axis ;

— positive $(+)$ and negative $(-)$ signs indicating that the graph is above and below the x-axis respectively ;

— the zeros of the function, which are the x-intercepts of the graph of $y = f(x)$, and the roots of the equation $f(x) = 0$;

— values of x where the graph is undefined.

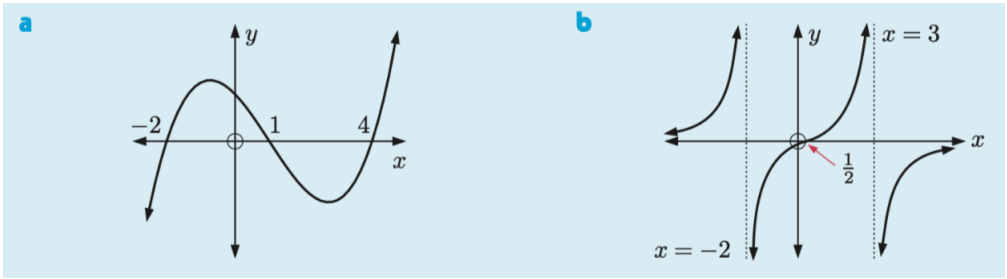
Example

Function	$y = (x+2)(x-1)$	$y = -2(x-1)^2$	$y = \frac{4}{x}$
Graph			
Sign diagram			

Remark. $\overset{\vdots}{0}$ A hollow dot or an open circle on the sign diagram indicates that a function is undefined at $x = 0$.

Example

Draw sign diagrams for :



Example

Draw a sign diagram for :

1. $(x+3)(x-1)$
2. $2(2x+5)(3-x)$

Example

Draw a sign diagram for :

1. $12-3x^2$
2. $-4(x-3)^2$

Example

Draw a sign diagram for :

$$\frac{x-1}{2x+1}$$

V- Rational functions

Definition

When we divide a linear function by another linear function, the result is a **rational function**.

The rational functions we consider in this course can be written in the form

$$y=\frac{ax+b}{cx+d}.$$

Definition

Rational functions are characterised by asymptotes, which are lines the function gets closer and closer to but never reaches.

These functions have asymptotes which are horizontal and vertical.

Definition

A reciprocal function is a function of the form

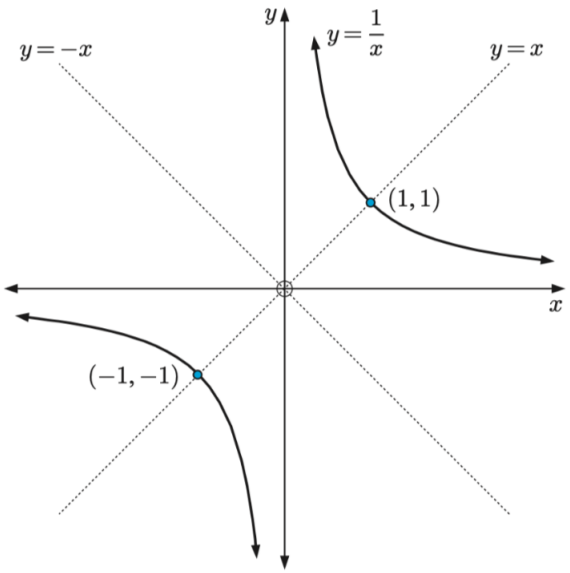
$$x\mapsto\frac{k}{x} \quad \text{or} \quad f(x)=\frac{k}{x}, \quad \text{where } k\neq 0.$$

Example

The simplest example of a reciprocal function is

$$f(x) = \frac{1}{x}.$$

The graph of $f(x) = \frac{1}{x}$ is called a rectangular hyperbola.



Notice that:

- $f(x) = \frac{1}{x}$ is undefined when $x = 0$
- The graph of $f(x) = \frac{1}{x}$ exists in the first and third quadrants only.
- $f(x) = \frac{1}{x}$ is symmetric about $y = x$ and $y = -x$.
- as $x \to \infty$, $f(x) \to 0$ (from above)
  as $x \to -\infty$, $f(x) \to 0$ (from below)
  as $x \to 0$ (from the right), $f(x) \to \infty$
  as $x \to 0$ (from the left), $f(x) \to -\infty$
- The **asymptotes** of $f(x) = \frac{1}{x}$ are the x-axis and the y-axis.

Property

To determine the equation of a horizontal asymptote, we consider the behaviour of the function as $|x| \to \infty$.

Investigation

1. Use the graphics calculator to examine the following functions for asymptotes :

   (a) $y = -1 + \frac{3}{x-2}$ (b) $y = \frac{3x+1}{x+2}$ (c) $y = \frac{2x-9}{3-x}$

2. State the domain of each of the functions in 1.

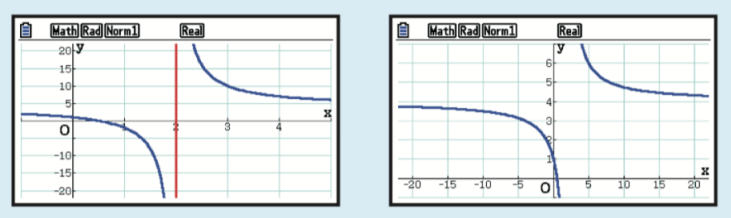
3. How can we tell directly from the function what its vertical asymptote is ?

Example

Consider the function

$$y = \frac{6}{x-2} + 4.$$

1. Find the asymptotes of the function.
2. Find the axes intercepts.
3. Use technology to help sketch the function, including the features from (a) and (b).



VI- Inverse functions

Definition

A function $y = f(x)$ *may or may not* have an inverse function.

If $y = f(x)$ has an inverse function, this new function :

— is denoted $f^{-1}(x)$;

— must indeed be a function, and so must satisfy the vertical line test ;

— is the reflection of $y = f(x)$ in the line $y = x$;

— satisfies $(f \circ f^{-1})(x) = x$ and $(f^{-1} \circ f)(x) = x$.

The function $y = x$, defined as $f : x \mapsto x$, is called the identity function.

Remark. f^{-1} is not the reciprocal of f.

$$f^{-1}(x) \neq \frac{1}{f(x)}.$$

Property

he domain of f^{-1} is equal to the range of f.

The range of f^{-1} is equal to the domain of f.

Example

$$f : y = 5x + 2 \quad \text{becomes} \quad f^{-1} : x = 5y + 2.$$

Example

Consider $f : x \mapsto 2x + 3$.

a. On the same axes, graph f and its inverse function f^{-1}.

b. Find $f^{-1}(x)$ using :

(i) coordinate geometry and the gradient of $y = f^{-1}(x)$ from (a) ;

(ii) variable interchange.

c. Check that $(f \circ f^{-1})(x) = (f^{-1} \circ f)(x) = x$.

Definition

Any function which has an inverse, and whose graph is symmetrical about the line $y = x$, is a self-inverse function.

Example

The reciprocal function

$$f(x) = \frac{1}{x}, \quad x \neq 0,$$

is said to be a self-inverse function as $f = f^{-1}$.

This is because the graph of $y = \frac{1}{x}$ is symmetrical about the line $y = x$.